Sanjay Adhikary

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PhD - Transportation systems Engineering
Department of Civil Engineering
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Career Objective

Motivated transportation researcher with a background in multimodal systems, last-mile accessibility, mode choice modelling, and environmental assessment. Aspiring to support sustainable cities and electric mobility programs through analytical research, evidence-based policy inputs, and effective communication of technical insights to diverse stakeholders.

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
PhD	IISc. Bangalore	Ongoing	2026-Present
M.Tech	IISc. Bangalore	7.5	2024-2026
M.Tech	IIT Guwahati	8.04	2020-2022
B.Tech	JGEC, Jalpaiguri	7.61	2015-2019
Senior Secondary	WBCHSE	86.6%	2015
Secondary	WBBSE	89.27%	2013

Experience

- **Uttar Banga Krishi Viswavidyalaya (UBKV)** *Dec 2022 - May 2023*
  Served as a guest lecturer in Civil Engineering Guest Lecturer
  – Delivered lectures and supervised labs for core civil engineering subjects (Surveying, Structural Analysis, Soil Mechanics).
  – Conducted evaluations, including exam preparation, grading, and practical examinations.

Projects

- **First and Last Mile fleet sizing for a multi-modal interchange** *Aug 2025 - May 2026*
  Under the guidance of Dr. Tarun Rambha,CiSTUP,IISc Bangalore M.Tech Thesis project
  – An agent-based simulation framework in AnyLogic for sizing first- and last-mile auto-rickshaw fleets at Yeshwantpur Railway Station, capturing train-arrival demand surges, luggage, group travel, waiting time and train delays within a multinomial-logit mode choice model. Scenario tests across fleet sizes and doubled bus frequency show that waiting time is the key lever, which makes auto fleet sizing and bus scheduling joint decisions.

- **An Exact Algorithm for the Multiple Vehicle Pickup and Delivery Problem** *Nov 2025 - Dec 2025*
  Under the guidance of Dr. Tarun Rambha,CiSTUP,IISc Bangalore M.Tech course project
  – Implemented a full integer programming framework for the Pickup and Delivery Problem (PDP) using Gurobi and the F1 formulation.
  – Integrated valid inequalities (transfer, adjacent, and pairing prior) to strengthen the LP relaxation, reducing the integrality gap and improving solver efficiency.
  – Evaluated model performance using Solomon's R-type and C-type datasets, achieving an optimal Z_{OPT} for instances up to 15 customers within a 1-hour time limit.

- **Unsupervised Learning for Cyber Intrusion Detection in Power Substations** *OCT 2025 - NOV 2025*
  Under the guidance of Dr. Puneet Rathore, CPS,IISc Bangalore M.Tech course project
  – Developed an unsupervised intrusion detection system (IDS) to identify malicious GOOSE traffic (Injection, Masquerade, Replay) in IEC-61850 substations.
  – Evaluated 7 models (K-Means, DBSCAN, Isolation Forest, GMM); identified OCSVM as the top performer with an F1-score of 83.4% and a 0.52 ms inference latency.

- **Swiss Metro Impact on Modal Share** *Jan 2025 - April 2025*
  Under the guidance of Prof.Abdul Rawoof Pinjari, Dept. of Civil Engineering,IISc Bangalore M.Tech course project
  – Estimated MNL, Nested Logit, and Mixed Multinomial Logit models using stated-preference survey data to analyze travel behavior.
  – Conducted elasticity and willingness-to-pay (WTP) analysis to derive policy insights for sustainable transport scenarios.

Technical Skills

- **Programming**: Python, R, JAVA (for AnyLogic), Matlab(Basic)
- **Transportation Engineering**: VISSIM, Gurobi, WinJuLEA, QGIS, AnyLogic
- **Data & Reporting**: MS Excel, Power BI, LaTeX, Origin Pro

** Elementary proficiency*